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IARCO RESEARCH PROPOSAL

RESEARCH TOPIC

Within-Session Productivity Trajectories in Students:

Rise–Peak–Decline Patterns and Peak Timing

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Introduction

Productivity of individual study sessions may not always be constant. A study session of sustained cognitive load can influence attention, focus, mental exhaustion, and task performance. Understanding these factors and how they vary over time might give us information that is not captured by an output measurement system.

A rise–peak–decline (RPD) pattern describes a measured variable that initially increases, reaches a maximum, and then decreases. Existing research on student productivity has emphasized aggregate outcomes such as academic achievement and self-control [1].

The primary knowledge gap is not only about if the students get tired when studying. It's also how productivity varies. With whether those variations follow discernible temporal patterns too. There are three linked questions that are important. First, in naturalistic study environments, the existence and organization of RPD type of productivity trajectories among students have not been described. Second, one universal "optimal" point in a study session has not been assumed with actual proof because the period of peak productivity may be different for individuals. Third, variations in trajectory shape or peak time may be linked to aspects of the study session like study intensity.

If we consider productivity as a within-session trajectory, the suggested study might be able to close this gap. In order to compare the start, middle, and end of sessions with varying total lengths, study observations will be mapped into a normalised session position. The same participants' repeated measurements will help us to examine both individual variance and general temporal patterns.

Central Research Question

This study tries to answer this question, "How does student productivity change within a study session, and how much does the intensity of the study session affect the timing and form of these within-session trajectories?" To answer this question, three further research questions are proposed:

1. (R1) During naturalistic study sessions, will students display consistent within-session productivity trajectories, such as rise-peak-decline (RPD) patterns?
2. (R2) How much does peak timing differ among students and sessions, and at what proportionate point does productivity peak?
3. (R3) Do variations in the timing, size, or form of within-session productivity trajectories correlate with study session intensity?

Sessions of various durations will be compared on a common temporal scale using Normalised Session Position (NSP), which is the percentage of elapsed session time relative to total session duration. This will help the study to differentiate between the relative location of an observation inside a session and the total study period.

Literature Review

The impact of task engagement on performance has been the subject of ongoing research on persistent cognitive activity. Mackworth's early vigilance studies highlighted the importance of time on task in sustained attention and showed that extended visual-search activity can have an effect on performance [2]. Successive studies suggest that sustained attention requires mental effort. It is also associated with stress and resource demands [3]. People have thought of mental tiredness as the ongoing energy and motivational costs of maintaining performance, with changes in how much effort and reward are valued which may possibly be affecting task engagement [4]. Mental fatigue has been linked in experimental research to changes in executive control, like decreased adaptability and ineffective planning for cognitively demanding activities [5]. Ackerman and Kanfer found that although the mean performance was higher for the longest test condition, subjective tiredness increased with longer time-on-task [6]. This might mean that extended testing does not always mean a drop in performance. These findings imply that weariness and performance may be impacted by ongoing cognitive effort. More importantly, the relationship between performance and elapsed time might be more complex than a simple decline.

Instead of looking at how to make study sessions better for each person, research in education is found to be more focused on study habits, study time, and academic outcomes. According to Plant et al., total study time alone was a weak predictor of college GPA [7]. According to Credé and Kuncel's large meta-analysis, study habits, skills, and attitudes were better predictors of college success than standard admissions tests and past grades[8]. Researchers on self-regulated learning also found that students' subjective choices about how to and when to study may not necessarily reflect how well they are doing[9]. Sievertsen et al. found systematic differences in test performance associated with school-day testing timing in Danish public schools [10]. This suggests that cognitive fatigue can affect student performance. Educational studies mostly focus on aggregate study behaviour, quality, academic outcomes, or performance at different times of day. They don't show how a student's productivity changes over a self-directed study session. Rise-and-decline patterns are a general sort of temporal dynamics that Raulo et al. identify across phenomena and areas [11]. This framework describes a variable that rises, peaks, and falls without assuming every observation follows the same path. However, rise and decline patterns in other situations do not prove student productivity follows an RPD trajectory. While educational research shows that study quality, habits, and contextual factors impact academic success [7]-[10], continuous cognitive load can impact attention, fatigue, and performance [2]–[6]. It is also unclear whether student productivity varies systematically throughout a study session, whether a peak can be found, how consistently that peak occurs among students and sessions, and whether study-session intensity is related to trajectory shape or peak time. So, rather than focusing on a single measure of study time or academic achievement, this study looks at student productivity as a dynamic within-session process.

Methodology

1. Research Design:
In this study, secondary school students will be recording repeated productivity assessments over many naturalistic study sessions using a prospective, longitudinal, repeated-measures observational design. This approach will allow repeated observations within sessions to characterise productivity trajectories (R1). It will also allow different students and sessions to have different study session intensities, which can be used to study how they relate to trajectory shape and peak timing (R3).

2. Participants:
After institutional approval, 15–18 years old secondary school students from one school will participate. A target of 30 participants will be set because this number is easy to recruit and each participant is expected to contribute about 12 sessions over four weeks. But 36 will be recruited for possible dropouts or incomplete participation.
Students who regularly complete 60-minute individual academic study sessions and who have a smartphone for study session logging will be included. Not being able to finish repeated measures or refusing to give informed consent or assent will be criteria for exclusion. After getting institutional approval, Participants will be recruited through the school and participation is voluntary.

3. Study Session Definition:
A continuous period of self-directed academic work on one or more academic activities which may last 60 minutes will be considered a study session. A break of more than 10 minutes will end the session, and any further study time will be noted as a new session.
The activity will stay in the same session and the topic type will be noted if a student switches subjects without taking a pause longer than ten minutes.

4. Productivity Measures:
For characterising a temporal trajectory, a session will be deemed legitimate if its entire time is recorded and it includes at least three valid productivity measurements.
At each measurement point, a ‘1–5 anchored scale’ will be used to rate productivity. Participants will rate how effectively they are working at that moment. 1 denoting extremely low productivity or the inability to maintain effective study, 3 denoting moderate productivity with some effective work, and 5 denoting very high productivity with sustained effective work.
Each task type will be having one objective output metric recorded, such as words written for writing assignments, pages completed for reading-based activity, or problems solved for mathematics. Instead of free-text answers, the output will be entered as a numerical value in a specific field.
Instead of comparing the objective output of fundamentally different activities, it will be compared within participants and within the same task type. Entries in which the standardised rating and output (within participant and task type) differ by more than 2 SD will be retained but flagged for a data-quality check.

5. Normalised Session Position:

The Normalised Session Position (NSP) will be calculated as follows:

> NSP (%) = (minutes elapsed at measurement ÷ total session length in minutes) x 100.

NSP = 0% will represent the baseline at the beginning of the session, and NSP = 100% will represent the final measurement at the ending of the recorded session. NSP will set up study sessions of different lengths on a comparable relative time scale so productivity trends across sessions of different lengths could be compared.

6. Timetable for Measurement:
A short mobile-based digital form will be given to participants to record data. An automated reminder will be set to minimise missing observations. Productivity ratings will be logged at the beginning of the session, every 30 minutes after that, and at the end of the session. A scheduled log falling within 10 minutes of the end will be replaced by the end-of-session rating. The 30-minute period will give finer resolution than with an hour at an acceptable burden.

7. Covariates and Study Intensity:
The main indicator of study-session intensity will be the planned session duration in minutes that is noted at the start of each session. When possible, perceived effort will also be included as an additional measure of intensity.
At the start of the study session, prior night sleep will be documented as self-reported hours of sleep. A simple 1–5 self-report scale will be used to record daily reported stress; higher values indicate more perceived stress. Sleep and stress will not be considered primary outcomes, but rather contextual factors.

8. Data Quality Rules:
In order to avoid replacing unobserved measurement numbers with estimated values, missing measurements will be reported rather than imputed. If a measurement is submitted within 10 minutes of the planned prompt, it will be deemed timely.
Participants who exhibit 80% or more similar productivity ratings throughout their recorded measurements will be examined in accordance with the predetermined data-quality standards and highlighted for potential satisficing behavior.
Participants who contribute fewer than five legitimate study sessions will not be included in participant-level trajectory analyses; exclusion criteria will be determined before analysis. Every exclusion will be recorded and reported rationale behind it.

9. Plan of Analysis:
To see within session trends and student variation, productivity will be plotted against Normalised Session Position (NSP) first for both individual participants and the entire sample.
A mixed-effects model will be used to estimate changes in productivity throughout NSP for (R1). Participants will be included as a random effect, and study sessions will be nested inside participants when the data supports it. For trajectory curvature, a quadratic NSP term will be included. An RPD-like trajectory requires a negative quadratic term (95% CI excluding 0) with the estimated peak between 10% and 90%

NSP. Without it, RQ1 has no clear answer criterion.
For (R2), the NSP at which predicted productivity is highest will be estimated per session and participant. Then, to describe how consistently peak productivity occurs across sessions and students, the mean, variability, and distribution of peak NSP values will be provided.
To find out if changes in planned session length or perceived effort are connected to changes in productivity trajectories, intensity × NSP and intensity × NSP² interaction terms will be tested, and peak NSP will also be examined as a function of intensity for (R3).
In order to ascertain if these variables are linked to variations in within-session productivity change, prior-night sleep and perceived stress will be included as covariates, and interactions with NSP will be investigated where supported by the data.

In addition to the relevant descriptive statistics and visualisations, the results will be presented using effect estimates with 95% confidence intervals. R will be used for statistical analysis.

Expected Results

The purpose of this study is to find out how student productivity evolves during naturalistic study sessions and how these trajectories differ with study-session intensity. Whether productivity exhibits recognisable within-session patterns like RPD-like trajectories, gradual decline, relatively stable trajectories, or mixed patterns across students and sessions will be determined by the analysis. This study will measure the difference in peak timing between students and sessions and be able to determine the location of maximum productivity on the 0–100% Normalised Session Position (NSP) scale. Plus it will also investigate whether variations in peak timing, trajectory shape, or post-peak change magnitude are related to study session intensity. We'll look at perceived stress and sleep duration to see if they can explain some of the observed diversity in productivity trajectories. Importantly, the study will tell the difference between evidence of a consistent RPD pattern among students and evidence that productivity changes consistently with session position. Just because there is a temporal correlation, doesn't always mean that there is a universal productivity curve. The results will nevertheless provide information about individual variation and the viability and resolution of regularly assessing productivity within naturalistic research sessions even if no consistent trend is found.

Importance

The project will fill a knowledge gap on student productivity as a within-session trajectory rather than a total study time or academic outcome. NSP's methodology will allow comparison of sessions of different lengths on a common relative scale and produce complementary performance metrics when task-specific objective output is paired with subjective productivity ratings. If trends were to be detected, the results may help schedule study time around productivity timing fluctuations. Further studies using controlled experimental methodologies, larger sample sizes,

or more objective performance indicators may build on the findings.

Ethics

3 All participants and, if under 18, their parents or legal guardians will give written informed consent before data collection. Volunteers can withdraw at any moment without penalty or academic consequence. Participants will be identified by anonymous research codes, and data will be saved in a password protected file for the researcher. People who take part in the study won't have to make any changes to how they normally study but only need to record short scores while doing so. This makes the study observational and low-risk. Before recruiting, school management approval is required.

Limitations and Mitigation

Limitation	Mitigation
Self-reported productivity ratings could be influenced by personal bias or changes in response patterns.	An anchored 1–5 scale and objective output measures will be used for cross-checking, with substantial divergence flagged (Methodology 4).
Repeated logging may lead to loss of data or missed measurements.	A recruitment buffer, short fixed measurement schedule, and predefined compliance rules will be used (Methodology 2, 6, 8).
Participants may provide repetitive ratings without careful engagement.	Repetitive-response patterns will be flagged and handled using predefined data-quality criteria (Methodology 8).
Rapid productivity changes between measurements may be missed.	Measurements at the selected 30-minute interval will balance temporal resolution with participant burden (Methodology 6).
Productivity may vary because of sleep, stress, time of day, or task type rather than session position.	These factors will be recorded as covariates, with task type recorded per entry and within-person mixed-effects analysis used to account for variation (Methodology 7, 9).
A small, self-selected sample from one school limits generalizability, and the observational design cannot establish causation.	Findings will be framed as descriptive and associational, giving a basis for larger and more controlled future studies (Methodology 1, 2, 9).

Conclusion

Student productivity might vary greatly within a study session, making total study duration insufficient to assess student performance over time. However, the shape, timing, and relationship of these within-session productivity variations with study-

session intensity are unknown. How productivity changes between naturalistic research sessions will be examined using prospective repeated measures, Normalised Session Position, and mixed-effects analysis. The findings may help explain productivity timing disparities and improve evidence-based study scheduling studies.

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